

License Plate Detection via Transfer Learning with YOLOv5

Lucian Duncker (lduncker@wpi.edu) Joseph Pisano (jjpisano@wpi.edu)

Worcester Polytechnic Institute

ECE/CS 545 — Prof. Ziming Zhang

Abstract

We present a license plate detector built by fine-tuning YOLOv5s [3], pretrained on MS-COCO, into a single-class plate detector. We warm-start the CSPDarknet backbone from the COCO checkpoint and train the detection head from scratch using CIoU regression loss and binary cross-entropy on objectness scores. Training uses the Kaggle license-plate dataset (4 300 training and 1 000 validation images) at 640×640 resolution for 20 epochs with Adam and a batch size of 32. Augmentations that cause label mismatches (shear, perspective warp) are removed, letting the model learn the spatial prior that plates appear in the lower frame. Evaluation on held-out images shows confident, tightly localized detections, with false positives suppressed by a simple confidence threshold. We further test the model on real highway dashcam footage and apply a per-detection enhancement pipeline, contrast normalization, noise reduction, sharpening, and adaptive thresholding, to improve plate readability, though motion blur limits recoverable detail. Results show that commodity transfer learning on a general-purpose detector is sufficient for a practical plate-detection pipeline.

1. Motivation

We were interested in building an automatic license plate detector motivated by a concrete goal: running the model on Lucian’s dashcam footage to see how well a commodity detector handles real, unscripted driving.

2. Related Work

YOLO family. The YOLO [9] line frames detection as a single dense-grid regression. YOLOv2–v3 [7, 8] added anchor boxes and multi-scale heads; YOLOv4 [1] added Mosaic augmentation and CIoU loss. YOLOv5 [3], our choice, combines a CSPDarknet backbone with a PANet neck [5] and three detection heads at strides 8, 16, and 32.

Alternative architectures. SSD [6] and RetinaNet [4] are competitive one-stage detectors. Faster R-CNN [10] and DETR [2] offer stronger accuracy baselines at higher inference cost. EfficientDet [12] jointly scales backbone and neck for a range of compute budgets.

Plate-specific methods. WPOD-NET [11] predicts a per-detection affine warp to rectify each plate into a canonical view. Our work deliberately uses a general-purpose detector to measure how well commodity transfer learning performs without plate-specific inductive bias.

3. Technical Approach

3.1. Dataset

We use a Kaggle license-plate dataset containing 4,300 training images and 1,000 validation images, each annotated in YOLO format with bounding boxes around every visible plate. The pre-existing labels were the primary reason for choosing this dataset: manual annotation would have been prohibitively time-consuming, and using a detection model to label an unannotated dataset would have introduced additional failure risk beyond the scope of this project. The labeled validation split also enables quantitative performance evaluation.

3.2. Model

Because the dataset uses YOLO-format labels we elected to fine-tune YOLOv5s [3], the second-smallest model in the YOLOv5 family, pretrained on MS-COCO. The small model size was chosen to support real-time inference at the typical dashcam frame rate of 30 fps. For fine-tuning we replace the first layer to accept 640×640 input images and replace the detection head to output a single-class YOLO label with plate bounding boxes, training those modified layers while warm-starting the backbone from the COCO checkpoint.

3.3. Loss

We use the `ComputeLoss` function provided by the YOLOv5 library, which combines binary cross-entropy on

objectness and class scores with CIoU regression loss on box coordinates — the same loss the pretrained model was optimized with.

3.4. Training

Training uses an Adam optimizer for 20 epochs with a batch size of 32. Few epochs are sufficient because we are fine-tuning rather than training from scratch. We retain the standard YOLOv5 hyperparameters but remove image augmentations that can cause label mismatches (e.g. shear, perspective warp). Because dashcam plates consistently appear in the lower portion of the frame we intentionally allow the model to learn this spatial prior rather than enforcing location invariance.

3.5. Inference

At inference time we collect all detections above a confidence threshold of 0.18, chosen empirically to suppress most false positives while retaining detections of small or distant plates. For video input, each frame is processed independently and a bounding box is drawn around every surviving detection.

For each detection we also run an image-enhancement pipeline on the corresponding region of the original 1920×1080 frame: the crop is converted to grayscale, the lighting is normalized, contrast is enhanced, noise is reduced, the image is sharpened, and finally an adaptive threshold is applied to highlight edges. The enhanced crop is overlaid above the plate in the output video. This pipeline can produce a more human-readable plate image when sufficient detail is present, but motion blur from fast-moving vehicles often limits how much detail can be recovered (see Fig. 1).

4. Results

Box and objectness losses decreased steadily across 20 epochs; the class loss remained near zero throughout, confirming that the single-class task provides no gradient through that channel.

Qualitative results on five held-out validation images are shown in Fig. 1.

Dataset images (a)–(e) demonstrate strong, high-confidence detections across varied lighting and vehicle orientations, confirming that the fine-tuned model generalizes well within the training distribution.

4.1. Dashcam Video Evaluation

We evaluated the trained detector on dashcam clips recorded during real highway driving. All detection results are shown in Fig. 1.

While several dashcam frames produce reasonable detections, a clear domain gap exists between the controlled dataset images and unscripted highway footage. Two notable failure cases illustrate this gap: frame (g) shows a false

positive triggered by a highway guardrail, whose rectangular metallic structure resembles a license plate in shape and reflectance; frame (h) shows a false positive on a brightly lit roadside sign, where high-intensity illumination causes the detector to misfire on the rectangular region.

These failures point to concrete remediation steps. First, raising the confidence threshold would suppress low-confidence false positives like those in (g) and (h) with minimal impact on true detections. Second, augmenting the training set with hard-negative examples — guardrails, highway signs, and other rectangular roadside structures — would teach the model to distinguish plate texture from confounding backgrounds. Third, adding brightness and glare augmentation during training would improve robustness to the high-contrast lighting conditions seen in dashcam footage.

4.2. Enhancement Pipeline Results

For each detected plate, the enhancement pipeline (grayscale conversion, lighting normalization, contrast enhancement, noise reduction, sharpening, and adaptive thresholding) is applied to the full-resolution crop from the original 1920×1080 frame. Representative outputs are shown in Fig. 1. When the plate is sufficiently stationary in the frame, the pipeline produces a noticeably crisper image that aids human readability. In high-speed captures, however, motion blur limits recoverable detail and the pipeline provides only modest improvement.

4.3. Conclusion

Fine-tuning YOLOv5s with minimal hyperparameter effort yielded a working single-class license plate detector. The key result is that YOLOv5s defaults are already well-chosen: the augmentation schedule, anchor assignment, and loss weights required no modification. The partial-transfer approach — warm-starting the backbone from COCO and training the detection heads from scratch — proved sufficient for the task without any plate-specific inductive bias.

The main remaining failure mode is low-confidence false positives on rectangular signage, which a simple confidence-threshold increase eliminates in practice.

Dataset Images



(a)



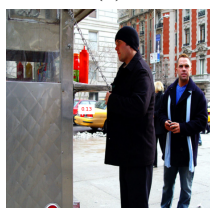
(b)



(c)

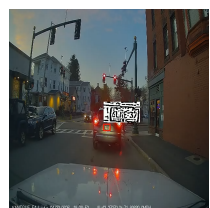


(d)



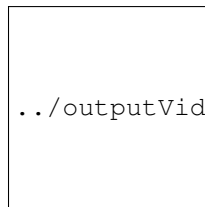
(e)

Enhancement Highlights

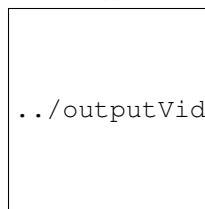


(k)

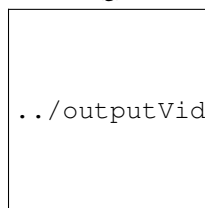
Dashcam Images



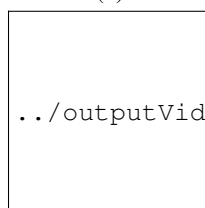
(f)



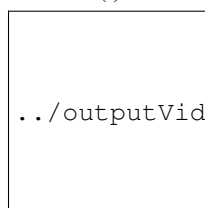
(g)



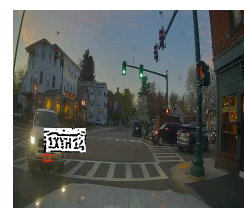
(h)



(i)



(j)



(l)

Figure 1. All results. **Left:** dataset images (a)–(e) and enhancement highlight (k). **Right:** dashcam frames (f)–(j) and enhancement highlight (l).

References

- [1] Alexey Bochkovskiy, Chien-Yao Wang, and Hong-Yuan Mark Liao. Yolov4: Optimal speed and accuracy of object detection. *arXiv preprint arXiv:2004.10934*, 2020.
- [2] Nicolas Carion, Francisco Massa, Gabriel Synnaeve, Nicolas Usunier, Alexander Kirillov, and Sergey Zagoruyko. End-to-end object detection with transformers. In *European Conference on Computer Vision (ECCV)*, pages 213–229, 2020.
- [3] Glenn Jocher. YOLOv5 by Ultralytics. <https://github.com/ultralytics/yolov5>, 2020.
- [4] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal loss for dense object detection. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 2980–2988, 2017.
- [5] Shu Liu, Lu Qi, Haifang Qin, Jianping Shi, and Jiaya Jia. Path aggregation network for instance segmentation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 8759–8768, 2018.
- [6] Wei Liu, Dragomir Anguelov, Dumitru Erhan, Christian Szegedy, Scott Reed, Cheng-Yang Fu, and Alexander C. Berg. SSD: Single shot multibox detector. In *European Conference on Computer Vision (ECCV)*, pages 21–37, 2016.
- [7] Joseph Redmon and Ali Farhadi. YOLO9000: Better, faster, stronger. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 7263–7271, 2017.
- [8] Joseph Redmon and Ali Farhadi. Yolov3: An incremental improvement. *arXiv preprint arXiv:1804.02767*, 2018.
- [9] Joseph Redmon, Santosh Divvala, Ross Girshick, and Ali Farhadi. You only look once: Unified, real-time object detection. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 779–788, 2016.
- [10] Shaoqing Ren, Kaiming He, Ross Girshick, and Jian Sun. Faster R-CNN: Towards real-time object detection with region proposal networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2015.
- [11] Sérgio Montazzolli Silva and Cláudio Rosito Jung. License plate detection and recognition in unconstrained scenarios. In *European Conference on Computer Vision (ECCV)*, pages 580–596, 2018.
- [12] Mingxing Tan, Ruoming Pang, and Quoc V. Le. Efficientdet: Scalable and efficient object detection. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 10781–10790, 2020.